

A Literature Review on Stock Price Prediction by Integrating Social Media Sentiment and Market Indicators Using Deep Learning Approaches

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Abstract

This paper reviews how social media and news sentiment can be integrated with market indicators to improve stock prediction, with emphasis on deep learning and multimodal fusion. A systematic review is conducted on literature from 2010 to 2024 across IEEE Xplore, ACM Digital Library, Scopus, and arXiv using explicit inclusion/exclusion criteria and a PRISMA-style screening procedure; foundational methods predating this window are also referenced where appropriate. To organize diverse findings, a four-dimensional taxonomy is proposed covering data sources, sentiment construction methods, fusion strategies, and predictive architectures. A comparative synthesis of representative studies yields three consistent patterns: multimodal approaches outperform unimodal (sentiment-only or price-only) baselines; hybrid fusion provides the most reliable balance between robustness and cross-modal interaction; and architectures coupling finance-specific encoders (e.g., FinBERT) with sequence models (e.g., LSTM or transformer-based forecasters) achieve strong error reductions, particularly in high-volatility regimes. Limitations persist—noisy and asynchronously timed sentiment signals, heterogeneous performance across markets and regimes, inconsistent evaluation protocols that hinder comparability, and complex yet opaque models that challenge adoption. In response, a practical evaluation checklist is provided and concrete future directions are outlined, including attention-based alignment, domain-specific pretraining across languages/markets, reinforcement learning for decision-making, cross-market robustness testing, and explainability.

CCS Concepts

• **Applied computing** → Operations research; Forecasting.

Keywords

Social media sentiment, Stock prediction, Deep learning, Multimodal fusion, FinBERT, LSTM, Transformer (Temporal Fusion Transformer)

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1 Introduction

Financial market forecasting is a longstanding challenge in both academic research and industry practice. The volatility, nonlinearity, and regime-dependent nature of stock markets complicate accurate prediction, limiting the effectiveness of traditional statistical and econometric models such as ARIMA and GARCH. These methods often fail to capture sudden shocks and the influence of non-financial factors, prompting a shift toward alternative data sources and machine learning approaches. Financial news tone contains predictive content for near-term returns, with high media pessimism forecasting downward pressure followed by reversion [1]. Large-scale microblog evidence further suggests that aggregate Twitter mood is predictive of the DJIA [2].

In particular, the rise of social media platforms has provided an abundant source of real-time investor sentiment. However, these data sources are inherently noisy, context-dependent, and asynchronous with market trading sessions, creating challenges for alignment and interpretation. Beyond news and social media, Google Trends-based nowcasting shows that simple models using query volumes can estimate contemporaneous activity such as initial unemployment claims, automobile sales, tourism, and housing starts [3].

However, lexicon-based methods lack the ability to capture contextual nuances such as sarcasm or domain-specific polarity shifts. Machine learning classifiers improved granularity, but only with heavy feature engineering and annotated corpora. Recent deep learning approaches, including LSTM and Transformer-based models, along with domain-specific variants such as FinBERT, have achieved substantial improvements in financial sentiment extraction [4]. These methods allow for context-aware embeddings that better reflect financial semantics, enabling more accurate downstream prediction. Domain-specific lexicons are necessary because generic negative lists misclassify a large share of finance terms, introducing noise into tone measures [5]. Building on this, domain-specific pretraining (FinBERT) has demonstrated strong performance on financial sentiment and information-extraction tasks in large financial corpora, supporting its adoption as a modern baseline [4].

On large S&P 500 panels, sequence models such as LSTM outperform memory-free baselines with statistically and economically significant gains, making LSTM a strong reference architecture [6]. Attention-based Temporal Fusion Transformers further combine multi-horizon accuracy with model-intrinsic interpretability via variable selection and self-attention [7]. Integrating sentiment features with market indicators has led to the emergence of multimodal frameworks. Recent multimodal deep-fusion designs combine a FinBERT-based text encoder with an LSTM price branch and report superior accuracy and trading performance relative to single-modality or simple concatenation baselines [8], with additional evidence under volatile regimes and with FinBERT-LSTM hybrids [9] [10]. Empirical evidence suggests that hybrid fusion, which balances modality-specific learning with cross-modal interaction, achieves the most consistent gains [9]. Predictive architectures that combine FinBERT encoders with sequence models (e.g., LSTM/Transformer) report strong performance across datasets [7] [9] [11]. Nevertheless, performance remains sensitive to market regimes, data alignment, and model reproducibility [9].

Despite these advances, current research still faces several shortcomings. Most existing surveys are narrative and fragmented, lacking a systematic review framework to consolidate findings. Results are difficult to compare due to heterogeneous datasets, inconsistent evaluation metrics, and varying baselines. Moreover, interpretability remains under-reported in finance applications, despite available architectures that explicitly provide attribution insights [7]. Prior empirical work underscores the need for standardized comparisons and significance testing (e.g., Diebold–Mariano) to ensure reported gains are meaningful and reproducible [6] [11] [12]. Moreover, reproducibility is often limited by proprietary data and unavailable code, hindering cumulative progress. These gaps underscore the need for a structured taxonomy and standardized evaluation practices.

This paper addresses these gaps through the following contributions: (i) It proposes a four-dimensional taxonomy of multimodal stock prediction approaches, spanning data sources, sentiment analysis methods, fusion strategies, and predictive architectures. (ii) It synthesizes empirical findings into comparative evidence, highlighting consistent performance trends across studies. (iii) It critically analyzes key challenges and limitations related to data quality, market heterogeneity, reproducibility, and model interpretability. (iv) It provides an evaluation checklist and concrete future research directions to standardize protocols and inspire further advances.

The remainder of the paper is organized as follows. Section 2 presents the review protocol. Section 3 introduces the taxonomy. Section 4 compares empirical findings. Section 5 discusses challenges. Section 6 proposes the evaluation checklist and future directions. Section 7 concludes.

2 Methodology (Systematic Review Protocol)

To ensure transparency and academic rigor, this paper adopts a Systematic Literature Review (SLR) protocol rather than a selective or narrative approach. The review followed a structured process including database selection, search strategy, inclusion and exclusion criteria, and a PRISMA-inspired screening procedure.

2.1 Databases and Timeframe

Four major academic databases—IEEE Xplore, ACM Digital Library, Scopus, and arXiv—were searched, covering the period from 2010 to 2024. This range captures the rise of social media platforms (e.g., Twitter, Reddit) and the increasing application of deep learning in financial forecasting.

2.2 Search Strategy

The search combined three categories of terms:

(i) Social media and sentiment: “Twitter”, “Reddit”, “FinBERT”, “sentiment”, “opinion mining”; (ii) Finance-related: “stock”, “market return”, “DJIA”, “financial forecasting”; (iii) Deep learning: “deep learning”, “LSTM”, “Transformer”, “multimodal fusion”.

The terms were connected with Boolean operators, for example: (“Twitter” OR “sentiment”) AND (“stock” OR “market”) AND (“LSTM” OR “Transformer”).

2.3 Inclusion Criteria

Studies were included if they satisfied several conditions: they examined financial market forecasting tasks, incorporated social media or financial news sentiment as input features, employed machine learning or deep learning methods, and were available in full text in English.

2.4 Exclusion Criteria

Excluded studies were those that focused only on sentiment analysis without forecasting, originated from non-academic sources, lacked full-text availability, or duplicated other entries (in such cases, the journal version was prioritized).

2.5 Screening Process

Following the PRISMA guideline, all retrieved records were first deduplicated, then screened by title and abstract, and finally assessed through full-text reading. Only studies meeting the above criteria were retained. The final set of included papers is consistent with the scope and page limit of this conference paper, ensuring a representative yet concise coverage.

2.6 Closing Statement

This systematic protocol guarantees that the review process is transparent, replicable, and academically sound, forming a reliable foundation for the subsequent taxonomy and comparative analysis.

3 Taxonomy of Multimodal Stock Prediction Approaches

To organize the diverse and fragmented literature, this paper proposes a four-dimensional taxonomy that classifies prior work on multimodal stock prediction (Figure 1). The taxonomy comprises four key facets: (i) data sources, (ii) sentiment analysis methods, (iii) fusion strategies, and (iv) predictive architectures. This framework provides a structured lens to interpret existing studies and represents one of the main contributions of this review. Prior studies highlight diverse signals and methods, from Twitter mood predicting stock indices to news tone analysis, underscoring the need for a structured taxonomy [1] [2].

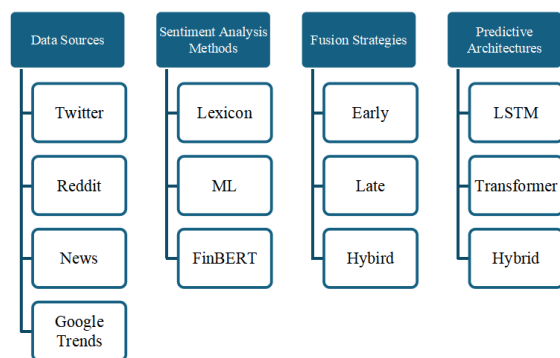


Figure 1: Taxonomy of multimodal stock prediction approaches

3.1 Data Sources

Sentiment signals for financial forecasting are typically derived from social media and financial news. Twitter is the most widely used source, reflecting high-frequency mood shifts among retail investors [2], while Reddit and other forums provide community-driven sentiment with longer discussions. Professional sentiment is often captured from news headlines and financial reports [1], which tend to be more reliable but lower frequency. Some studies also employ Google Trends to proxy public attention toward financial assets [3].

The strength of social media data lies in timeliness and scale, but they are noisy, context-dependent, and susceptible to manipulation. In contrast, financial news offers more domain relevance but may lag in capturing rapid market shifts. Effective multimodal approaches often combine these heterogeneous sources to balance immediacy and credibility. Google search–query indices are widely used for nowcasting real-time economic indicators (e.g., unemployment claims, auto sales), providing “predicting-the-present” signals when official releases lag [3].

3.2 Sentiment Analysis Methods

Approaches to extracting sentiment have evolved considerably. Early studies relied on lexicon-based methods such as the Loughran–McDonald financial dictionary [5], which assign polarity scores to predefined terms. These methods are transparent and interpretable but fail to capture contextual nuances such as negation or sarcasm.

Early supervised classifiers (e.g., SVM or logistic regression with bag-of-words/n-gram features) offered finer granularity than lexicon methods [5] but required heavy feature engineering and sizeable labeled corpora; domain-specific pretrained language models—most notably FinBERT—subsequently surpassed such baselines on financial texts by capturing contextual, finance-specific semantics [4]. Despite their accuracy, deep models require significant computational resources and remain challenged by domain adaptation across regions or asset classes.

3.3 Fusion Strategies

Early fusion concatenates text and numerical features at the input level [13]. Late fusion aggregates modality-specific predictors to improve robustness, at the cost of reduced cross-modal interaction [13]. Hybrid/deep fusion learns modality-specific encoders and merges them at an intermediate layer, preserving per-modality structure while enabling cross-modal learning [13]. In finance, deep-fusion architectures—e.g., a FinBERT text branch paired with an LSTM price branch—consistently outperform single-modality and simple concatenation baselines in both prediction accuracy and portfolio backtests [8]. Under high-uncertainty regimes, dual-stream attention/hybrid variants tend to be strongest [9] [10]. However, this strategy increases model complexity, training cost, and data requirements.

3.4 Predictive Architectures

LSTM remains a strong baseline for financial time-series forecasting [6]. Transformer-based forecasters—especially the Temporal Fusion Transformer (TFT)—deliver competitive long-horizon accuracy with intrinsic interpretability [7]. In multimodal settings, hybrid/deep-fusion architectures that pair FinBERT text encoders with LSTM price branches achieve consistent gains over single-modality models or naïve concatenation [8], with additional improvements reported for FinBERT–LSTM dual-stream/attention designs under high-volatility regimes [9] [10].

While hybrid designs show promise, they remain sensitive to hyperparameter tuning and often require large datasets for stable training. Moreover, robustness across different market regimes and scalability in real-time settings are open challenges.

This four-dimensional taxonomy provides a coherent framework for classifying and comparing prior research on multimodal stock prediction. By organizing studies along data sources, sentiment analysis methods, fusion strategies, and predictive architectures, it offers a structured lens through which to interpret the literature. This taxonomy also lays the foundation for the comparative evidence presented in Section 4.

4 Comparative Evidence and Empirical Findings

Although prior studies share the goal of improving stock prediction with sentiment and deep learning, they differ widely in datasets, sentiment sources, and evaluation metrics. As a result, direct comparison across all models is difficult. Instead, this section synthesizes evidence to highlight consistent trends, with representative studies summarized in Table 1.

4.1 Sentiment-only vs Multimodal

Early sentiment-only approaches demonstrated non-trivial predictive content but with limited robustness. Classic evidence shows that aggregate Twitter mood correlates with broad market indices such as the DJIA, indicating that collective emotions can anticipate short-run market movements [2]. More recent cross-market analyses find that news-based sentiment generally exerts a stronger and more persistent impact on returns than social-media sentiment, with effects varying by market and context [14]. These results, together with practical issues of noise and asynchrony in social

Table 1: Representative Empirical Studies on Sentiment-informed Stock Prediction.

Study (Year)	Data	Sentiment Method	Fusion Strategy	Predictive Model	Metrics	Key Findings
Bollen et al. (2010) ^[2]	Twitter mood + DJIA	Aggregate sentiment indices	–	Correlation models	Accuracy, Correlation	Twitter mood correlated with DJIA (accuracy up to 87.6%), early proof of sentiment value.
Loughran & McDonald (2011) ^[5]	10-K filings	Lexicon (LM dictionary)	–	Baseline	Polarity index	Domain-specific lexicon outperformed generic dictionaries; widely used baseline.
Chen et al. (2024) ^[8]	Prices + Technical indicators + News headlines	FinBERT	Hybrid / Deep fusion	FinBERT + LSTM	MAE, MAPE; portfolio backtest (returns)	Deep-fusion outperforms single-modality and simple concatenation baselines; improves prediction error and simulated trading performance.
Bacco et al. (2024) ^[9]	Bank stocks; Twitter + prices	Twitter sentiment (FinBERT)	Hybrid (dual LSTM + attention)	Two-stream LSTM	DA	In 2022 high-volatility, multimodal > historical-only; regional differences observed.
Gu et al. (2024) ^[10]	Prices + FinBERT sentiment	FinBERT	Hybrid (FinBERT+LSTM)	LSTM	MAE, MAPE	Reported 37.5% MAPE reduction vs LSTM, 79.5% vs DNN; strong multimodal gains.
Halder (2022) ^[11]	NASDAQ-100	FinBERT	Hybrid (FinBERT+LSTM)	FinBERT-LSTM	MAE, MAPE	Reduced MAPE by 3.24% vs LSTM, 20.2% vs MLP; FinBERT improved robustness.
Nyakurukwa & Seetharam (2024) ^[14]	News + Social (multi-market)	Lexicon/ML	–	Returns / Regression	Returns, t-stats	Cross-market evidence: news-based sentiment generally exerts a stronger and more rational impact on stock returns than social-media sentiment; effects vary by market.
Mao et al. (2011) ^[17]	Survey, News, Twitter, Search	Mixed sources	Early fusion	Regression	Adjusted R ²	Adding sentiment/search data improved explanatory power by 117% (relative).
Kirtac &	Financial	LLM sentiment	–	Long–short	Backtest	LLM sentiment

streams, suggest that sentiment-only models often struggle to generalize across regimes.

By contrast, multimodal approaches—especially deep-fusion models coupling a FinBERT text branch with an LSTM price branch—consistently outperform sentiment-only or price-only baselines across datasets [8] [9] [10]. Subsequent work under high-uncertainty conditions shows that hybrid (dual-stream) architectures with attention better balance modality-specific representation and cross-modal interaction, yielding superior robustness relative to unimodal or simple late-fusion alternatives [9]. In particular, finance-specific encoders (e.g., FinBERT) paired with sequence models (e.g., LSTM/Transformer) consistently reduce error metrics versus standalone LSTM or MLP baselines across diverse datasets [10] [11]. Overall, the comparative evidence indicates that while sentiment-only models can capture broad correlations in calmer periods [2], hybrid multimodal designs generally outperform across markets and especially in turbulent regimes [9].

4.2 Fusion Strategies

Comparisons across fusion strategies show that early fusion is simple but prone to noise propagation [13]. Late fusion improves robustness by aggregating independent modality-specific predictions but misses fine-grained interactions [13]. In finance, hybrid/deep fusion generally performs best: FinBERT + LSTM deep-fusion outperforms single-modality and simple concatenation baselines [8], and dual-stream attention architectures are strongest under volatile regimes [9] [10]. Nevertheless, hybrid designs increase computational cost and data requirements, which can limit practical deployment.

4.3 Predictive Architectures

LSTM remains a strong baseline for financial time-series forecasting [6]. In multimodal settings, deep-fusion models that pair a FinBERT text encoder with an LSTM price branch deliver consistent gains over single-modality or naïve concatenation approaches [8] [10]. Transformer-family forecasters (e.g., TFT) further improve long-horizon accuracy with model-intrinsic interpretability, and hybrid LSTM–Transformer variants are increasingly competitive [7]. Under high-uncertainty or highly volatile regimes, dual-stream/attention hybrids tend to be most robust [9] [10]. These advantages come at the cost of higher computational budgets and careful hyperparameter tuning.

In summary, consistent empirical patterns emerge: (i) multimodal models outperform sentiment-only approaches, (ii) hybrid fusion strategies provide the best trade-off between accuracy and interaction modeling, and (iii) hybrid architectures combining FinBERT with LSTM/Transformer deliver strong results across reported datasets. These findings establish the empirical foundation for the next section on challenges and limitations in multimodal stock prediction.

5 Challenges and Limitations

Despite encouraging progress, existing studies on sentiment-informed stock prediction face several challenges that limit their

reliability, reproducibility, and generalizability. Early studies already noted that while sentiment correlated with indices, its explanatory power was modest and prone to noise [2]. This section discusses four major categories of limitations: (i) data quality and alignment, (ii) market heterogeneity and regime dependence, (iii) methodological issues and reproducibility, and (iv) model complexity and interpretability.

5.1 Data Quality and Alignment

Social media data are inherently noisy, containing spam, sarcasm, and misleading information that distort sentiment signals. Classification accuracy deteriorated during volatile regimes, indicating alignment is challenging [9]. Bollen et al. [2] demonstrated correlations between Twitter mood and DJIA movements, but the explanatory power remained limited due to data volatility. Recent cross-market evidence further shows news sentiment tends to have larger and more persistent effects than social-media sentiment, underscoring source heterogeneity [14]. Temporal alignment further complicates analysis, as tweets occur continuously while markets operate in fixed sessions. Aligning sentiment features with trading hours or price windows remains an open challenge, reducing the robustness of empirical findings.

5.2 Market Heterogeneity and Regime Dependence

Another limitation arises from inconsistent performance across markets and regimes. Cross-market experiments reveal that models may succeed in one region but fail elsewhere, reflecting substantial market heterogeneity [9]. Bacco et al. [9] showed that multimodal LSTM models improved predictions for North American banks but had weaker gains in European markets, indicating regional heterogeneity. Likewise, Halder [11] reported that FinBERT-LSTM models improved predictive accuracy on NASDAQ-100 stocks but faced declining robustness during high-volatility regimes. These findings suggest that many models overfit to specific datasets and fail to generalize across markets or economic conditions, limiting their practical applicability.

5.3 Methodological Issues and Reproducibility

Comparative evaluation is hindered by a lack of standardized protocols. Reported improvements are often dataset-specific, making it difficult to compare gains across studies without common benchmarks [10]. Few studies apply statistical forecast comparison tests such as Diebold–Mariano, limiting confidence in claimed improvements [11]. Studies use heterogeneous datasets, rolling windows, and performance metrics (e.g., MAE, MAPE, DA), making results difficult to compare directly. For instance, Gu et al. [10] demonstrated significant improvements with a FinBERT-LSTM framework, while Bacco et al. [9] emphasized dual-stream architectures under high uncertainty. However, differences in datasets and evaluation setups prevent meaningful cross-study benchmarking. Moreover, few studies provide open-source code or share complete datasets, limiting reproducibility and hindering cumulative scientific progress.

5.4 Model Complexity and Interpretability

The most accurate models often involve complex hybrid architectures that combine transformers with sequence models. Hybrid FinBERT-LSTM models reduce error metrics but substantially increase computational demands, limiting their accessibility for practitioners [11]. Interpretability remains under-explored, even though architectures such as TFT offer attribution insights that could enhance trust in predictions [7]. Halder [11] showed that FinBERT-LSTM significantly reduced forecasting error compared to baselines, but at the cost of increased model complexity and computational requirements. Similarly, hybrid architectures described by Gu et al. [10] demand large datasets and extensive tuning. While these models achieve state-of-the-art accuracy, their “black-box” nature hinders interpretability, which is essential for adoption in financial practice where transparency and accountability are critical.

5.5 Noise-Handling Methods: Comparative Effectiveness

Prior research widely acknowledges that social-media sentiment is contaminated by spam/bots, sarcasm/irony, off-topic drift, and timing mismatches with market trading windows, which together weaken predictive content and cross-study comparability [2] [14]. In financial text, domain-specific semantics further complicate polarity assignment (e.g., “liability” or “short”), motivating finance-tuned language models to mitigate lexical/semantic noise at source [4]. To address the reviewer’s request for a comparative perspective, this subsection consolidates the effectiveness and trade-offs of the main families of noise-handling methods used in sentiment pipelines for stock prediction.

At the account/content level, spam and coordinated inauthentic activity are most efficiently reduced by bot-oriented screening that exploits network, temporal, and metadata regularities; such approaches consistently improve the stability of aggregate sentiment features by excising a small but high-impact fraction of posts and accounts [15]. Sarcasm and irony pose a different challenge—surface polarity intentionally diverges from intended meaning. Contextual models trained specifically for sarcasm detection on Twitter-like text yield sizable gains in downstream sentiment fidelity and thus reduce error propagation into forecasting targets [16]. Complementary to these detectors, domain-specific encoders (e.g., FinBERT) trained on financial corpora attenuate misclassification arising from finance-specific vocabulary and contextual cues, giving cleaner sentence- and document-level signals for fusion with market indicators [4]. Because social posting is continuous whereas trading is session-bound, temporal alignment (restricting to market hours, validating lead-lag structure via cross-correlation, and z-score normalization on rolling windows) remains essential to curb asynchrony-induced noise that is particularly salient for social media relative to news [2] [14]. Finally, when aggregating post-level scores into asset-day features, robust statistics (e.g., medians or light Winsorization) and lightweight ensemble sentiment (combining a lexicon baseline with a finance-tuned encoder) typically produce the most reliable improvements per unit complexity, smoothing idiosyncratic errors without suppressing informative bursts around events.

For transparent evaluation, it is recommended reporting two ablations tied to Eqs. (1)–(3): a model with vs. without bot filtering (to quantify account-level sanitation) and single-scorer vs. ensemble sentiment (to quantify robustness to scorer-specific errors), under identical temporal alignment and leakage-aware splits. In practice, a minimal yet effective recipe is: account-level screening for bots/spam → market-hours restriction with validated lead-lag → optional sarcasm gating when labels are available → domain-specific encoding for sentiment extraction → robust aggregation and, where feasible, a simple ensemble. This combination directly targets the dominant noise channels documented by prior literature while preserving a reporting protocol that facilitates replication and cross-market comparison [4] [14] [15] [16].

5.6 Cross-market Mechanisms Behind North American–European Performance Gaps

The observed performance gaps between North American and European settings are consistent with differences in the linguistic, informational, and behavioral environments documented in prior work. First, language and domain shift likely play a central role: finance-tuned encoders such as FinBERT are trained predominantly on English financial corpora and mitigate semantic ambiguity in English texts (e.g., polarity for “liability” or “short”), but they do not directly address multilingual or mixed-register inputs that are more prevalent in European sources; this asymmetry can depress sentiment extraction quality outside English and propagate into downstream forecasts [4]. Second, timing frictions interact with social media’s around-the-clock posting to create market-specific asynchrony: social signals that peak outside local trading sessions are noisier and less aligned with return windows, a problem already noted for social streams relative to news and plausibly more acute when posting rhythms and exchange hours are mismatched across regions [2] [14]. Third, the relative strength of news versus social sentiment differs across markets: cross-market evidence shows news-based tone tends to be more “rational” and persistent than social-media sentiment, with effect sizes varying by market, implying that multimodal models whose gains rely on social inputs will benefit more in settings where retail participation and social penetration are higher—features more characteristic of U.S. equities than several European venues [14]. These considerations align with empirical reports that hybrid FinBERT–sequence architectures deliver larger improvements in North American datasets, while gains in European samples are weaker or require stronger alignment and tuning [9], and that robustness deteriorates in high-volatility regimes unless architectures and validation protocols are carefully adapted to the local information environment [11].

Taken together, these mechanisms suggest concrete avenues for reconciliation. Linguistic adaptation—via domain-specific encoders suited to the target language—should narrow gaps that stem from semantic noise in non-English sources [4]. Market-hours alignment and lead-lag validation can reduce asynchrony and improve the mapping from social signals to returns in markets where posting peaks are off-session [2] [14]. Finally, modality weighting that reflects the relative informativeness of news versus social cues across regions, combined with leakage-aware evaluation under local regimes, should yield more comparable performance profiles

and help explain why multimodal gains are systematically larger in North America than in Europe [9] [11] [14].

In summary, the literature faces four key challenges: (i) noisy and misaligned sentiment data, (ii) heterogeneous performance across regions and market regimes, (iii) methodological inconsistencies and limited reproducibility, and (iv) complex yet opaque model designs. These challenges highlight the need for standardized evaluation protocols, reproducible benchmarks, and interpretable model architectures. They also motivate the evaluation checklist and future research directions discussed in the next section.

6 Evaluation Checklist and Future Directions

The limitations discussed above underscore the need for standardized evaluation practices and clearer research roadmaps. To address inconsistencies in datasets, metrics, and reproducibility, this section outlines a practical evaluation checklist followed by key future research directions.

6.1 Evaluation Checklist

To ensure comparability and replicability across studies, future work should adhere to the following guidelines:

6.1.1 Data and Preprocessing. Clearly specify data sources (e.g., Twitter, Reddit, financial news) and document preprocessing steps such as spam removal, sarcasm handling, and temporal alignment. Bollen et al. [2] highlighted the noise inherent in Twitter data, underscoring the need for rigorous cleaning.

6.1.2 Sentiment Construction. Explicitly report the sentiment tools used, such as the Loughran–McDonald dictionary [5] or pretrained models like FinBERT [4]. This transparency enables meaningful comparisons between traditional lexicon-based baselines and advanced transformer-based approaches.

6.1.3 Experimental Protocol. Use temporal splits or rolling-window validation to avoid data leakage. Given evidence that the predictive content of social-media sentiment is weaker and less persistent than news-based sentiment in many markets [14], and that performance can deteriorate under high-uncertainty regimes [9], use leakage-aware temporal splits (e.g., rolling windows) and stress-tests across regimes.

6.1.4 Evaluation Metrics. Report multiple performance metrics, including MAE, MAPE, and Directional Accuracy (DA). Where possible, conduct statistical significance tests. Mao et al. [17] used Adjusted R^2 , while Halder [11] employed MAE/MAPE, showing the value of diverse but standardized reporting.

Standard definitions are used (y_i denote the target and $\hat{y}_i$ the predictions).

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (1)$$

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (\text{defined when } y_i \neq 0) \quad (2)$$

$$\text{DA} = \frac{100}{n-1} \sum_{i=2}^n \mathbb{1}[(y_i - y_{i-1})(\hat{y}_i - \hat{y}_{i-1}) > 0] \quad (3)$$

6.1.5 Baseline Comparisons. Always benchmark against established baselines such as ARIMA, SVM, or sentiment-only models. Include text-only and price-only baselines; a recent deep-fusion study shows that integrating FinBERT-based news sentiment with an LSTM price branch outperforms single-modality models and simple combinations [8].

6.1.6 Reproducibility. Provide code and data where feasible, or at least detailed documentation of experimental settings. Bacco et al. [9] emphasized cross-market differences but limited reproducibility due to proprietary datasets, highlighting the need for transparency.

6.2 Future Directions

6.2.1 Attention-based Multimodal Alignment. One promising avenue is the application of attention mechanisms to align asynchronous sentiment and market data. Transformer-based models such as FinBERT have shown strong performance in capturing contextual semantics [11], and Gu et al. [10] demonstrated the benefit of integrating FinBERT with LSTM predictors. Future work could extend this by using attention to dynamically weight sentiment signals relative to price movements, improving alignment and interpretability.

6.2.2 Domain-specific Pretrained Models. While FinBERT [11] represents an important step toward domain-specific sentiment modeling, it is trained primarily on English financial news. Expanding pretrained models across languages and markets could enhance generalization. For example, applying financial BERT variants to multi-regional corpora would address the regional heterogeneity highlighted by Bacco et al. [9]. Such efforts could reduce domain adaptation costs and improve cross-market performance.

6.2.3 Reinforcement Learning for Decision-making. Most existing studies focus on predictive accuracy rather than trading decisions. Reinforcement learning (RL) offers a framework for directly optimizing trading strategies using multimodal inputs. Incorporating sentiment signals into RL-based agents could bridge the gap between predictive modeling and actionable decision-making. Although not widely adopted yet, this direction has potential to transform sentiment-informed forecasting into practical portfolio management tools.

6.2.4 Cross-market and Regime Robustness. Future research should explicitly evaluate models under varying market regimes and across different regions. Bacco et al. [9] observed notable differences between North American and European markets, while Halder [11] reported declining robustness during high volatility. Establishing benchmarks that systematically test models under diverse regimes would ensure greater reliability and external validity.

6.2.5 Explainability and Trustworthiness. Finally, enhancing interpretability is critical for adoption in financial practice. Hybrid deep learning models (e.g., FinBERT-LSTM) provide accuracy gains [11] but are perceived as “black boxes.” Techniques such as attention weight visualization or post-hoc explanation methods could improve trustworthiness, ensuring that models are both accurate and understandable for practitioners.

In summary, the proposed checklist provides a set of minimum standards for evaluating multimodal sentiment-informed stock prediction, while the outlined future directions highlight promising avenues for advancing the field. Together, these recommendations aim to standardize evaluation and inspire robust, reproducible, and practically relevant research in multimodal stock prediction.

7 Conclusion

This review has examined the integration of social media sentiment and traditional market indicators in stock price prediction, with particular emphasis on the role of deep learning and multimodal data fusion. The topic is of growing importance as financial markets become increasingly influenced by digital platforms, and the ability to extract and align sentiment signals with market movements can enhance both academic understanding and practical decision-making.

The study systematically synthesized prior literature through a transparent review protocol, organized findings into a four-dimensional taxonomy, compared empirical results across representative studies, and critically assessed methodological challenges. The analysis demonstrates that multimodal approaches consistently outperform unimodal baselines, hybrid fusion strategies provide the most reliable balance between accuracy and interaction modeling, and architectures that combine FinBERT with temporal sequence models such as LSTM or Transformers deliver strong predictive performance on the reported datasets.

Beyond identifying empirical trends, the review highlights their significance. For scholars, the findings clarify which methodological choices yield the greatest improvements and where gaps remain in terms of evaluation protocols, reproducibility, and interpretability. For practitioners, the evidence suggests that combining structured and unstructured information offers tangible forecasting advantages, but implementation requires careful attention to robustness, computational cost, and explainability.

Nevertheless, important limitations persist. Data quality and alignment issues, regional heterogeneity, inconsistent evaluation standards, and the opacity of complex hybrid models all constrain generalizability and adoption. Future research should pursue standardized benchmarking frameworks, the development of domain-specific multilingual models, reinforcement learning approaches that translate predictive power into actionable trading strategies, and techniques to enhance interpretability and trustworthiness.

By addressing these gaps, future studies can move beyond isolated demonstrations of predictive accuracy toward a more cumulative, transparent, and practically relevant body of research. In doing

so, the field can advance both theoretical knowledge of financial sentiment dynamics and the design of more robust tools for market forecasting and portfolio management.

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